

MLFF-DATA8: MACE Artifacts, Replay, and Protocol Identity

mdstats project

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Status and scope

MLFF-DATA8 converts DATA5-DATA7 evidence into executable, fixed-file MACE job artifacts. It does not run training, choose a checkpoint, activate a locked test, or perform active learning. The first adapter is intentionally narrow and is locked to `mace-torch==0.3.16` with the `NATIVE_MACE_FIXED` exposure backend.

The stage owns five boundaries for consumers that still use the DATA8 fixed-file representation:

1. MACE-readable target extended XYZ plus compact sidecar provenance;
2. explicit atomic-reference mappings and target training weights;
3. replay train/monitor preparation with disjointness evidence;
4. complete, immutable `TrainingProtocolIdentity` records; and
5. independent final-development and cross-validation job directories in which held-out evaluation and locked interpolation test data never appear in a training configuration.

DATA8 is an artifact-preparation gate. DATA9 owns process execution, candidate checkpoint evaluation, replay-retention enforcement, out-of-fold aggregation, final committee construction, and protocol freeze for the consumers that still use this representation.

Restored current P5 is not authorized by this broad DATA8 protocol graph. Current post-selection cross-validation and fresh final production are governed by `mlff_post_selection_p5_spec.md` and descend from `PostSelectionMethodIdentity` -> `role policy` -> `role plan` -> `fitted preparation` -> `PostSelectionMaterialization` -> `run evidence`. `TrainingProtocolIdentity` and `Data8PreparationBundle` remain current only for separately current non-P5 consumers that still use them and as historical provenance; successful deserialization does not make them current P5 authority.

Why the adapter is version locked

MACE is an evolving training application rather than a static file format. The adapter therefore verifies the exact source behavior on which its artifacts depend. For MACE v0.3.16, the required behaviors are:

- the replay head `pt_head` is sorted before target heads;
- validation heads are evaluated in loader order;
- native scheduler, patience, and best-checkpoint logic use only the last validation head;
- target data can be duplicated internally when the target/replay ratio falls below `real_pt_data_ratio_threshold`;
- multi-head fine-tuning can replace the requested learning rate and EMA settings unless `force_mh_ft_lr` is true;
- target-head, distributed-sampler, and combined training loaders can discard the final partial batch through `drop_last`;
- `dry_run` and `save_all_checkpoints` are available; and
- external replay supports `pt_train_file` and `pt_valid_file`.

The broad DATA8 fixed-file adapter uses target-last ordering where that consumer's checkpoint contract requires native target-last scheduling, disables implicit duplication by writing `real_pt_data_ratio_threshold: 0.0`, requests all candidate checkpoints, and records a loader dry-run prediction. Restored P5 does not inherit DATA8's historical fold-local checkpoint-monitor topology; its exact common-monitor topology and mode-specific execution are owned by `mlff_post_selection_p5_spec.md`.

A source probe is evidence, not a claim that any future MACE version is compatible. Every supported version requires its own tested compatibility record.

Public contracts

MaceCompatibilityPolicy

Locks package name, package version, release tag, commit, official source URLs, and policy version. The initial policy accepts only `mace-torch==0.3.16`.

MaceSourceProbe

Records content digests and verified source semantics:

- replay-head ordering;
- target-last validation ordering;
- last-head checkpoint behavior;
- implicit target duplication;
- the multi-head loss, LR/EMA override, and target-loader truncation branches that the qualified runtime must control;
- dry-run support;
- save-all-checkpoint support; and
- fixed-file adapter acceptance.

The adapter fails closed if any required behavior is absent.

MaceCheckpointControlPolicy

For broad DATA8 consumers using this legacy/general fixed-file policy, the initial mode is `NATIVE_TARGET_LAST_WITH_EXTERNAL_CONSTRAINT_AUDIT`. It requires:

- target validation head last;
- all candidate checkpoints saved;
- native early stopping effectively neutralized by large patience;
- external replay-retention audit when replay is enabled; and
- no locked-test evidence in checkpoint selection.

Restored P5 checkpoint/adaptive-stop target evidence is instead the external campaign-common `M_mon` defined by the current P5 specification.

MaceLoaderDryRun

Predicts the realized loader contract before training:

- head and validation-head order;
- native checkpoint head;
- requested and effective target/replay counts;
- implicit target duplication factor;
- target/replay ratio; and
- fixed-file backend identity.

For target count N_{ft} , replay count N_{pt} , and threshold r , the v0.3.16 compatibility emulation repeats target data until the realized ratio satisfies the source behavior. DATA8 defaults to $r = 0$, so balancing is owned explicitly by mdstats rather than hidden inside MACE.

Qualified execution semantics

The current execution identity is pinned to `mace-torch==0.3.16` and the source-qualified mdstats MACE wrapper. Loss realization is **method-specific**.

For separately accepted weighted paths represented through DATA8, including P3 target-size screening and the accepted P5-scratch method where applicable, the parser-facing configuration explicitly carries the weighted loss and all objective coefficients. Replay-enabled weighted configurations also carry:

```
loss = "stress"
force_mh_ft_lr = true
real_pt_data_ratio_threshold = 0.0
```

Every ordinary one-head parser-facing configuration explicitly carries `multiheads_finetuning = false`; replay configurations carry `multiheads_finetuning = true` where that current consumer uses the DATA8 multi-head path. This prevents MACE 0.3.16 parser defaults from silently changing an accepted mode.

The historical wrapper mutation that replaced MACE's multi-head `UniversalLoss` with `WeightedEnergyForcesStressLoss` belongs only to the weighted DATA8 method family. It **must not** be applied to restored foundation P5. Current `naive_fine_tuning` and `multihead_replay` foundation P5 resolve native `UniversalLoss` with the fixed D2 parameters through the P5 method owner and existing MACE seam. There is no global cross-mode MACE loss-family authority.

For target-size execution, the same authenticated target-size authority activates complete target-head and combined-loader coverage: `drop_last` is false, the realized batch count is $\text{ceil}(N / B)$, every exported target `frame_uid` is present exactly once, and target-size distributed sampler paths are rejected unless separately qualified. No frame is duplicated to fill a partial batch. The resolved evidence is attached to the existing TRAIN2 runtime summary; missing, stale, or mismatched evidence cannot authorize a restart or downstream current artifact.

Restored foundation P5 has different accepted exposure semantics (`drop_last=true` and single-process, with replay/`pt_head` first then target for multihead replay) and is governed by `mlff_post_selection_p5_spec.md` rather than this P3 complete-batch clause.

MaceExtxyzArtifact

Stores one verified target data file and its sidecar manifest. Every exported configuration contains the minimum MACE-readable keys:

```
REF_energy
REF_forces
REF_stress
config_weight
config_energy_weight
config_forces_weight
config_stress_weight
frame_uid
```

The exporter performs an ASE write/read round trip and verifies frame order, labels, weights, and stress representation. Per-atom floating columns are written with at least 17 significant decimal digits. The ASE 3.29 default `%16.8f` format is not used because it can round Cartesian positions and force labels by several nanounits and violate the lossless DATA8 contract.

For restored foundation P5, `config_weight` may remain a neutral transport field but is not an active loss/identity layer; binary property masks remain active.

Executable loss family and weighting layers

For separately accepted **weighted** methods represented through DATA8, the generated MACE configuration emits the resolved objective explicitly:

```
loss = "stress"
energy_weight
forces_weight
stress_weight
```

`loss = "stress"` selects pinned MACE's `WeightedEnergyForcesStressLoss`. Its native reductions consume `config_weight` (`ref.weight`) and the per-frame property weights linearly and apply the global coefficients exactly once, outside those reductions. A weighted path may not rely on MACE's `forces_weight = 100` default.

The old blanket statement that `UniversalLoss` is forbidden on every current path is retired. Foundation P5 intentionally uses native `UniversalLoss` under accepted D1/D2. That is valid precisely because foundation P5 no longer claims the weighted-path `config_weight` contract. Do not emulate foundation `UniversalLoss` with a patched loss, square-root weighting trick, residual pre-scaling, sample duplication, or a second `mdstats` loss engine.

For methods that consume them, the three weighting layers remain distinct:

Layer	Owner	Exported as
global loss coefficients	[objective] TrainingObjectivePolicy	/ energy_weight, forces_weight, stress_weight config keys
per-configuration weight	[weighting] ConfigurationWeightPolicy	/ config_weight
local property weights	canonical label presence	config_energy_weight, config_forces_weight, config_stress_weight

Local property weights are availability masks - 1.0 when the canonical label is present, 0.0 when it is absent - and SHALL NOT carry per-frame copies of the global coefficient ratio.

The [objective] and [weighting] policy readers validate raw configuration and current-schema values before canonicalization. Global coefficients are finite nonnegative reals with at least one positive value; the configuration equalization flag is an actual boolean; configuration multipliers and bounds are finite positive reals satisfying the normalized-mean constraint; and focus collections contain only declared string or positive-integer elements. Malformed current-schema values SHALL fail rather than become valid policy identity through int, float, or bool coercion. Historical representations remain admissible only through an explicit supported compatibility reader.

TrainingObjectivePolicy owns the global component coefficients only; it SHALL NOT carry a loss-family field. Loss family belongs to the applicable method identity. Foundation P5 therefore resolves its fixed UniversalLoss objective through PostSelectionMethodIdentity, while weighted P3/scratch consumers keep their existing objective/weighting owners.

Stress contract

REF_stress is an ASE six-component stress vector in eV/Angstrom³ with order

xx yy zz yz xz xy

and ASE's stress sign convention. Stress and virial are never conflated. A missing stress label is represented through zero stress loss weight, not by a fabricated physical value.

The six-component transport representation does not redefine the restored foundation-P5 UniversalLoss reduction, which is governed by D2 and consumes all nine stored Cartesian stress entries after the dependency's representation conversion.

Atomic-reference mapping

DATA7 AtomicReferenceFitRecord values are serialized into the target head as an explicit mapping from atomic number to energy. Conceptual record names are never placed in the MACE E0s field. The target head also carries an explicit head-local atomic_numbers literal containing only elements present in target configurations. For the DATA8 fixed-file job directories this stage owns, the top-level atomic_numbers remains the union of target and replay elements for model construction. This distinction is required by MACE 0.3.16: without the head-local table, the target head inherits replay-only elements and incorrectly demands target E0 values for them. The fit-record digest remains in the applicable job manifest and identity.

The union rule is not restored foundation-P5 authority. Current P5 materialization is governed by mlff_post_selection_p5_spec.md: its top-level and target-head atomic_numbers literal covers the target training/monitor elements that carry fitted residual E0s, and the executable model is reconstructed from the authenticated foundation checkpoint/head, so a replay-only species need not appear in that target-side literal to survive model reconstruction. DATA8 does not supply a second P5 model-construction policy.

Restored foundation-P5 atomic-reference fit/selected-head residual/transfer semantics are owned by mlff_post_selection_p5_spec.md; DATA8 transport does not create a second E0 solver or fit authority.

ReplayPreparationPlan

Supported modes are:

- NONE;
- PRESELECTED local replay;

- EXTERNAL_TRUE_LABEL local replay;
- EXTERNAL_PSEUDOLABEL local replay; and
- MP_SHORTCUT preparation-only planning.

Fixed-file execution requires local replay train and monitor artifacts. MP_SHORTCUT is rejected at job-bundle construction until its selected replay file has been materialized and inspected.

Replay train and monitor configurations must be geometry-disjoint. The monitor is never used for gradients and later provides retention evidence. Each file records its SHA-256 digest, frame count, element set, geometry identities, and property keys.

Current P5 replay-label default/legacy-ambiguity semantics are owned by the P5 specification and cannot be overridden by a broad DATA8 default.

ReplayRetentionPolicy

Declares the retention metric, maximum tolerated degradation, disjoint-monitor requirement, and failure behavior. DATA8 serializes this policy for consumers that use this representation; the execution owner evaluates it against saved checkpoints.

FoundationCheckpointIdentity

Binds protocol artifacts to the exact foundation checkpoint path, file digest, model label, and optional model metadata. A changed checkpoint creates a new applicable training identity.

TrainingProtocolIdentity

For separately current non-P5 DATA8 consumers, TrainingProtocolIdentity binds all choices represented by that broad protocol record that can change optimization or interpretation:

- target label domain;
- generic training mode;
- foundation checkpoint;
- replay-plan digest;
- DATA7 objective, weight, E0-fit, and checkpoint-policy digests;
- MACE compatibility lock and source probe;
- optimizer and precision settings;
- checkpoint-control policy;
- exposure backend;
- loader dry-run realization;
- selected training level; and
- random seed.

A naive protocol and a replay protocol represented through this family are different identities even if their target XYZ files are identical.

TrainingProtocolIdentity is **not current restored-P5 method authority**. Old DATA8 protocol records cannot authorize P5 cross-validation, final production, restart, or publication merely because their bytes deserialize.

SealedEvaluationArtifact

Records locked interpolation-test membership and lineage without writing a test XYZ file. The artifact is explicitly unmaterIALIZED. A downstream owner may activate it only after the applicable freeze boundary exists.

MaceJobArtifact

For consumers that still use the DATA8 fixed-file job family, MaceJobArtifact represents one final-development job or one independent cross-validation fold. It binds:

- target training and checkpoint-monitor artifacts;
- optional replay train and monitor artifacts;
- explicit fold evaluation artifact excluded from the training configuration;
- YAML configuration;
- run script;
- protocol identity;

- loader dry-run record; and
- file checksums/manifests.

This fold-local target-monitor shape is **not** current restored-P5 topology. Current P5 folds bind external common M_mon in their role plans.

Data8PreparationBundle

Collects all final and fold jobs for exactly one target label domain, plus local replay artifacts and sealed outer evaluation metadata, for consumers that still use this fixed-file representation. Multiple incompatible target label domains require separate DATA8 bundles.

Current restored P5 SHALL NOT be forced back through Data8PreparationBundle merely to preserve historical structure.

Extended-XYZ and sidecar split

Extended XYZ carries only training labels, weights/masks, and a stable frame_uid. Complete provenance remains in a canonical JSON sidecar keyed by frame UID:

- source occurrence and source-content identities;
- source frame index;
- composition and condition;
- geometry and label-payload identities;
- eligibility and selection lineage;
- applicable weight and E0-fit records; and
- file and policy digests.

This prevents long provenance payloads from becoming fragile XYZ header text.

Job layout

For the broad DATA8 fixed-file representation, a bundle has the conceptual layout:

```
data8_bundle/
  shared/
    replay/replay_train.xyz
    replay/replay_monitor.xyz
  jobs/
    final/
      target_train.xyz
      target_monitor.xyz
      mace_config.yaml
      run_mace.sh
      job_manifest.json
    fold_00/
      target_train.xyz
      target_monitor.xyz
      fold_evaluation.xyz
      mace_config.yaml
      run_mace.sh
      job_manifest.json
    ...
  data8_bundle.json
```

fold_evaluation.xyz is an evaluation artifact only. Its path must not appear in mace_config.yaml. No locked interpolation-test XYZ exists in this tree.

The presence of target_monitor.xyz in this DATA8 layout does not define current restored-P5 monitor ownership; P5 uses its external common-monitor plan lineage.

MACE configuration contract

For DATA8 consumers, the target head contains explicit property keys and E0 values. A replay job also declares pt_train_file and pt_valid_file. The applicable generated configuration includes the fields required by that method,

including foundation/model, mode/head definitions, property keys, optimizer, seed, device/precision, checkpoint retention, replay-threshold controls, and no test path.

Historical/current non-P5 DATA8 weighted configurations may include target/replay training-head scales and DATA7 objective weights where their accepted method consumes them. **Restored current P5 does not:** its target/replay training-head scalar weights are retired, and its foundation objective is fixed by current D2/P5 specification.

The first DATA8 adapter supports fixed-file training only. CUSTOM_EPOCH_RESAMPLE and MULTI_JOB_RESAMPLE remain separate backends and cannot be represented merely by writing one static YAML file.

Cross-validation jobs

The following nested-monitor DATA8 job description is retained only for separately current consumers/historical interpretation of this bundle family:

1. fitted features/E0/weights are fit only on the authorized gradient-training domain;
2. the DATA8 job's checkpoint-monitor artifact is exported as its validation file;
3. the held-out evaluation fold is exported separately and excluded from training configuration;
4. replay artifacts and optimizer/checkpoint rules match that protocol; and
5. execution evaluates held-out evidence only after checkpoint choice.

It does **not** define restored P5 CV. Current P5 fold membership is selected-only train + held-out outer evaluation + purge, with exact external common M_mon used for checkpoint/adaptive-stop control.

Failure rules

DATA8 fails closed for its consumers when:

- the MACE source probe does not match the active version lock;
- more than one target label domain enters one bundle;
- a required DATA7 fold is missing or has incompatible lineage;
- the foundation checkpoint cannot be hashed;
- an E0 mapping is incomplete for required target elements;
- a target label or required training weight/mask is unavailable;
- replay train and monitor overlap where forbidden;
- local replay is incomplete;
- MP_SHORTCUT is passed to fixed-file execution;
- a locked-test or fold-evaluation path enters a training configuration;
- an extended-XYZ round trip changes labels, stress, weights/masks, or frame order; or
- serialized digests or file checksums fail.

For restored P5, additional current failure rules in `mlff_post_selection_p5_spec.md` govern UniversalLoss, exact common monitor, selected-head residual transfer, exposure, currentness, and publication.

Non-goals

DATA8 does not execute MACE, inspect actual checkpoint files, enforce replay retention numerically, aggregate out-of-fold results, choose a final model, construct a committee, freeze a protocol, activate locked tests, calibrate uncertainty, or acquire active-learning labels. It also does not define current restored-P5 method identity, common-monitor topology, residual-transfer rule, or final-publication ordering.

References

1. MACE documentation, “Multihead fine-tuning,” official project documentation, accessed 2026-07-28.
2. MACE documentation, “Training,” official project documentation, accessed 2026-07-28.
3. ACESuit/MACE v0.3.16 source, `mace/cli/run_train.py`, official tagged source.
4. ACESuit/MACE v0.3.16 source, `mace/tools/train.py`, official tagged source.
5. ACESuit/MACE v0.3.16 source, `mace/tools/multihead_tools.py`, official tagged source.
6. ASE documentation, extended XYZ and stress conventions, version 3.29.0.

DATA9A hardening amendments

The production adapter applies the following stricter contracts where the DATA8 representation is still consumed:

- foundation fine-tuning records a checkpoint-bound `foundation_residual` atomic-reference fit; direct total-energy E0 fitting is a from-scratch-only fallback in the historical/general topology;
- the foundation checkpoint and local replay files are staged under `shared/`, YAML paths are relative, and run scripts change to their own directory;
- extended-XYZ verification compares species/order, PBC, cell, positions, energy, forces, stress, config type, and all weights numerically;
- replay inspection rejects nonfinite/misshaped labels and internal exact duplicates, records stress coverage, and binds pseudo-labels to a checkpoint;
- top-level `atomic_numbers` in DATA8 job directories is the union of target and replay elements (restored foundation P5 follows its own specification);
- `heads.target_head.atomic_numbers` is the target-only element set, preventing replay-only species from becoming target-head E0 requirements; and
- one explicit DATA7 ladder size is bound into every applicable protocol identity.

These amendments are implemented in 0.20.37a0 as part of DATA9A hardening. Current restored-P5 residual/transfer authority is the P5 specification, not this historical hardening clause.

DATA9A2 executable-serialization amendments

The v0.3.16 runtime contract is stricter than a native YAML interpretation. `atomic_numbers`, `heads`, every nested head `atomic_numbers` value, and every nested head E0s mapping are emitted as scalar strings containing deterministic Python literals. DATA8 does not emit unsupported `weight_pt` or `weight_ft` options.

The loss name recorded in this amendment was the lowercase parser choice `universal`. The subsequent DATA8 weighted topology standardized its current weighted consumers on `stress`; restored foundation P5 now intentionally uses native `UniversalLoss` again under the independently accepted P5 method. The historical transition note is retained so old evidence remains interpretable.

For preselected fixed-file replay, historical DATA9A2 target/replay training exposure was realized by multiplying each training structure's extended-XYZ `config_weight` by a corresponding target/replay head scale. Validation, monitor, fold-evaluation, and locked-test structures remained unscaled. The target sidecar recorded base, scale, and realized weights. These amendments are implemented in 0.20.39a0 and remain historical evidence for that topology. They are **not** current restored-P5 training-head weighting semantics.

Selectable fine-tuning precision

`MaceOptimizerPolicy.default_dtype` SHALL be either `float32` or `float64` and SHALL be serialized into every generated MACE configuration that uses this policy. Because the optimizer policy is part of `TrainingProtocolIdentity` for its consumers, changing precision creates a different broad protocol even when dataset artifacts are unchanged. For restored P5, precision identity is bound through the current P5 method/runtime lineage instead.

The MPA-0 foundation checkpoint may remain uniformly `float64`. A DATA8 bundle references it unchanged; runtime conversion to `float32` is requested through MACE's `default_dtype` option. DATA8 SHALL NOT claim output precision from configuration text alone. Runtime realization must inspect the saved model and target-head model and prove that all floating parameters and buffers are uniformly the requested dtype where that qualification is required.

DATA9A9b production orchestration

For the DATA9A9b topology, the orchestrator calls the native DATA8 builder only after all planned final and fold-local DATA7 bundles verify. Its production plan binds the foundation checkpoint, MACE compatibility probe, optimizer/checkpoint/export policies, selection size, and exact replay train/monitor plan before artifact generation. A `ProductionData8ArtifactRecord` binds the native DATA8 bundle digest and every relative file path and SHA-256 in the emitted tree. A partial or modified tree is not accepted as production materialization evidence.

This historical/general production topology does not override the current P5 plan/materialization graph.